

Demand Forecasting Competition

Organization: Crompton Greaves Consumer Electricals Ltd. (CGCEL)

Competition Judges: Anup Gholkar & Mridul Bhardwaj

Date: 30 April 2026

1. Project Objective

To develop an accurate demand forecasting model for CGCEL product categories such as **Fans, SDA, LDA & Pumps**, to improve inventory planning, production scheduling, and supply chain responsiveness

2. Business Need

CGCEL operates in a highly dynamic market where product demand is influenced by **seasonality, festive cycles, and weather conditions**. Improved forecasting is required to reduce stock-outs, minimize excess inventory, & support better decision-making across the supply chain

3. Scope

- Historical sales and demand analysis and generate forecasting at Segment-SKU level
- Identification of seasonal and regional demand patterns
- Model development, validation, and competition submission

4. Participants will receive as input

- Historical Sales Data: Fans, SDA, LDA & Pumps sales data at Branch-SKU level (monthly/weekly granularity)
- Product Master: SKU-level details including category, subcategory/segment classification and product attributes
- Geography Master: Region/Branch/State/City-level sales breakdown to capture regional demand patterns
- Festival/Holiday Calendar: Indian festival dates and national/regional holidays relevant to demand spikes
- Weather Data: Historical temperature and rainfall data mapped to branch offices

5. Participants are expected to deliver (Deliverables)

- Forecast Output File (in excel/csv): Predicted sales demand quantities at Branch-Segment & Branch-SKU level for the defined forecast horizon of 4 months Jan'26, Feb'26, Mar'26 & Apr'26, for all PLs
- Working Code / Notebook: Fully reproducible code covering data preprocessing, feature engineering, model training, validation, and prediction
- Approach Document: Detailed write-up of methodology, tools, assumptions, and key findings (see Section 8 for template)

Success Criteria:

- Forecasting model outperforms the provided baseline accuracy
- Demand seasonality and category trends are effectively captured
- Final submission is completed within competition timelines
- Recommendations are relevant and actionable for Crompton Greaves planning teams

6. Submission Guidelines

Parameter	Details
Submission Format	CSV file with columns: Year, Month Name, Region, SalesOffice Code, SalesOffice Nme, Segment, SKU_Code, Predicted_Demand
File Naming Convention	TeamName_Submission_v1.csv
Maximum Submissions Allowed	Up to 3 attempts (best score will be considered)
Submission Deadline	End of Week 2 as per project timeline
Submission Mode	Upload via competition portal / email to judges

Judging Criteria & Weightage:

#	CRITERIA	WEIGHTAGE
1	Forecast Accuracy (WAPE / RMSE / SHAP)	40%
2	Approach & Methodology (Feature engineering, model selection, validation)	25%
3	Code Quality & Reproducibility	15%
4	Business Relevance of Insights & Recommendations	10%
5	Approach Document & Presentation Quality	10%
	Total	100%

7. Tools & Technologies Allowed

Participants are free to use the following tools and technologies:

CATEGORY	ALLOWED TOOLS / EXAMPLES
PROGRAMMING LANGUAGES	Python, SQL
ML/STATISTICAL LIBRARIES	scikit-learn, XGBoost, LightGBM, statsmodels, ARIMA/SARIMA, etc.
DATA HANDLING	Pandas, NumPy, Excel
VISUALIZATION	Matplotlib, Seaborn, Plotly
DEVELOPMENT ENVIRONMENT	VS Code or any installed IDE system

> Note: Any tool or library used must be clearly documented in the Approach Document

8. Approach Document — Required Template

Every team must submit an Approach Document covering the following sections:

Approach Document Structure

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- | — 1. Executive Summary
- | Brief overview of the approach and results
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- | — 2. Data Understanding & EDA

	Key observations, distributions, missing values,
	outlier analysis, correlation insights
	└─ 3. Data Preprocessing & Cleaning
	Handling missing data, outliers, transformations,
	encoding techniques
	└─ 4. Feature Engineering
	└─ Seasonal / cyclical features
	└─ Weather-based features
	└─ Festival / holiday indicators
	└─ Lag features & rolling window aggregates
	└─ Any external features added
	└─ 5. Model Selection & Experimentation
	└─ List of models tried (with rationale)
	└─ Hyperparameter tuning approach
	└─ Comparison of model performances
	└─ Final model selection justification
	└─ 6. Validation Strategy
	Train-test split methodology, cross-validation,
	time-series specific validation (walk-forward, etc.)
	└─ 7. Results & Model Performance
	Final accuracy metrics (WAPE, RMSE, etc.),
	comparison with baseline
	└─ 8. Tools & Technologies Used
	Complete list of languages, libraries, platforms,
	and external data sources

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|— **9. Key Insights & Recommendations for CGCEL**

| **Actionable business recommendations based on**
| **demand patterns discovered**

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|— **10. Limitations & Future Improvements**

| **Known gaps, areas for improvement,**
| **what could be done with more time/data**

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|— **11. References**

Papers, blogs, documentation referred to

9. Risks & Mitigation

- **Poor data quality:** Early validation and cleansing
- **Demand volatility:** Category-level and regional segmentation
- **Seasonal spikes:** Inclusion of festival, and weather drivers
- **Overfitting:** Robust validation strategy (walk-forward validation, cross-validation)

10. Timeline

- All teams need to submit their deliverables in 2 weeks

11. General Information

- All the documents, data, program or any other information/file generated during this competition belongs to Crompton Greaves Consumer Electrical Limited (CGCEL)
- **Shared input data (historical sales, product masters, etc.) shouldn't be uploaded on any public platforms (Kaggle – including open-source AI platforms such as ChatGPT, Gemini, etc.)**